RealNucleus: Nuclear Binding as Dual Coulomb Confinement, without a Strong or Weak Force

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Mathematical formulation by the author; implementation developed with AI assistance (Claude, Anthropic).

Abstract

RealQM is a parameter-free, real-space model of quantum mechanics in which charge appears as a sum $\Psi = \sum_i \psi_i$ of one-particle components, each carrying unit charge on a non-overlapping domain separated by free boundaries, the ground state minimising the *Coulomb* energy over components and partition [1, 2, 3]. We carry this model into the **atomic nucleus** through the proton–electron picture, treating protons and electrons as the *same* object — equal-mass charge clouds differing only in the *sign* of the charge — interacting through **Coulomb forces only**, with **no strong force**. Binding is by **dual confinement**: the electron clouds hold the protons together against their repulsion while the protons cage the electrons against theirs. Calibrating one energy scale on the deuteron (2.22 MeV), the alpha emerges at ∼87% of its experimental binding with no fitting (alpha/deuteron ratio ≈11 versus 12.7; earlier single-mesh runs gave 13.1, but on a refined ladder the deuteron settles some 27% deeper and the ratio plateaus near 11, see §5.9), and the ladder ^{2}H $\to$ ^{4}He $\to$ ^{8}Be $\to$ ^{16}O binds as *single* connected units with near-constant binding per nucleon (saturation) as the nucleus adds shells. No *weak* force is needed either: the model "neutron" (1p + 1e) is computed *unbound*, so free-neutron β-decay into a proton and an electron follows from the same Coulomb energetics. Hydrogen burning then reduces to a Coulomb rearrangement of protons and electrons, 4H $\to$ ^{4}He + 2e, two electrons moving inside as the alpha's glue and two remaining as the atomic shell, with no proton-to-neutron conversion and hence no positrons or neutrinos required by the process (and none denied). The model is solved by signed Poisson and free-boundary evolution on a GPU and runs in a web browser. Throughout we keep one distinction strict: a nucleus's binding energy is a *measured* quantity — the mass deficit of independently weighed proton, neutron and nucleus via $E = mc^2$, with no nuclear model of any kind — whereas every number reported here is *computed* from the Coulomb model under a single calibration. What is directly measured are the three masses; the binding follows from $E = mc^2$ as their deficit, relative to the chosen decomposition into protons and neutrons. The comparison is thus a parameter-free *prediction* placed against a model-free mass-deficit *measurement*, on the single scale fixed by the deuteron.

Keywords: nuclear binding; proton–electron model; Coulomb confinement; real-space methods; multiphase wave functions; free-boundary problems; alpha clustering; strong force; weak force; beta decay.

1 Introduction

For ninety years it has been a fixed point of physics that the atomic nucleus cannot be held together by electromagnetism. The protons are all positively charged and repel; something else, a short-range *strong* force — mesons in Yukawa's formulation, residual QCD today — must overcome that repulsion. This force is one of the four fundamental interactions, and the whole of nuclear physics since Chadwick's neutron in 1932 rests on it. To propose that a nucleus is bound by Coulomb forces *alone*, with no strong force at all, therefore looks not merely wrong but impossible: it is the textbook reason the strong force had to be invented.

This article nonetheless asks exactly that question — and reports a computation that, to our own surprise, answers it in the affirmative for the light nuclei. The move that reopens the question is to abandon the point-particle, protons-and-neutrons description and return to the older *proton–electron picture* of the nucleus, but with protons and electrons represented as *charge clouds* in the manner of RealQM. We then ask: does the ordinary Coulomb interaction, with no separate strong force, already account for nuclear binding?

The proton–electron picture takes a neutron to be a bound proton–electron pair ($n = p + e$), consistent with β-decay $n \to p + e + \bar{\nu}$. A nucleus of mass number A and atomic number Z then consists of A protons and $A - Z$ electrons. The deuteron ($^2\mathrm{H} = \mathrm{p} + \mathrm{n}$) is $2\mathrm{p} + 1\mathrm{e}$; the alpha ($^4\mathrm{He} = 2\mathrm{p} + 2\mathrm{n}$) is $4\mathrm{p} + 2\mathrm{e}$; oxygen-16 is $16\mathrm{p} + 8\mathrm{e}$.

In RealQM [1, 3] charge is carried by one-particle components on *non-overlapping* spatial domains, with continuity and Bernoulli (pressure-balance) conditions across the free boundaries between them. Non-overlap takes the place of antisymmetry and the exclusion principle; there is no determinant and no spin. We apply this model to the nucleus with one essential symmetrisation: protons and electrons are treated as the same kind of object — equal-mass charge clouds — so that the only distinction between them is the sign of their charge. The interaction is Coulomb and nothing else.

The mechanism that emerges we call **dual confinement**. A single electron placed between two protons binds them: its negative cloud sits in the region where it most lowers the energy, between the two positive clouds, and holds them against their mutual repulsion — this is the deuteron, and it is the same mechanism that binds the one-electron molecular ion H_2^+. Reciprocally, the two protons confine the electron, which would otherwise spread away. In a larger nucleus the same balance operates on both species at once: the electron clouds keep the protons from flying off, and the proton clouds keep the electron clouds together. The nucleus is a self-confined two-component Coulomb system, stabilised by the RealQM non-overlap condition.

We report that this purely electromagnetic packing reproduces the scale and trends of nuclear binding. The central claim of the paper is therefore the title's: **nuclear binding is dual Coulomb confinement of proton and electron charge clouds, and no strong force is needed**, neither for the binding itself nor for the saturation (constant binding per nucleon) of larger nuclei.

1.1 The terms of the test: measured binding versus computed binding

Before any result, we fix the epistemic ground on which the whole paper stands, because it is what gives the comparison its force. **A nucleus's binding energy is a measured quantity, not a theoretical one.** One weighs the proton, the neutron, and the nucleus — three independent masses, each obtained by cyclotron-frequency comparison of a single trapped ion in a Penning trap, to parts in 10^{10}–10^{11} — and forms the mass deficit

$$B \;=\; \big(Z\, m_p + N\, m_n - m_{\text{nucleus}}\big)\, c^2. \tag{1}$$

No strong force, no shell model, no liquid drop, no fitted parameter, and no model of this paper enters (1). The single physical law it invokes, $E = mc^2$, is itself verified directly to $\sim 10^{-7}$ (Rainville *et al.*, 2005), and the deficit is cross-checked *as an energy* wherever the same quantity is measured independently — the deuteron's 2.2246 MeV deficit is also its measured capture-γ energy; heavier nuclei give it as photodisintegration thresholds and reaction Q-values. The binding energy is therefore an **empirical quantity** — not a direct reading, but the mass deficit of three weighed masses, converted by $E = mc^2$ and referred to the chosen proton–neutron decomposition — owing nothing to any nuclear model.

RealNucleus does the categorically opposite thing. It *sets up a model* — unit charge clouds on non-overlapping domains, interacting by Coulomb forces alone — and **computes** a binding energy, measuring nothing about the nucleus itself. Once a single energy scale is fixed on the deuteron (2.22 MeV), every further number below is a *prediction*.

This asymmetry — a **computed prediction** on one side, a **weighed fact** on the other — is what makes the test both fair and unforgiving, and we ask the reader to hold it throughout. The model cannot hide behind theory: nothing is fitted to obtain the agreement, and the benchmark owes nothing to a nuclear model, so the comparison stands in the open on both sides — there is neither a tuned parameter to absorb a departure nor a rival model to dispute one. The same measured benchmark that certifies the parameter-free agreement on the alpha-conjugate nuclei (^{4}He, ^{12}C, ^{16}O) also exposes, without appeal, where the present model is structurally undetermined — the odd, open-shell nuclei, whose computed binding depends on an assumed geometry (Sec. 5.8). We regard this exposure as a feature of the method, not a defect of the presentation: a parameter-free computation placed against a model-free measurement is the strongest form a nuclear-binding claim can take, and the weakest to conceal.

2 The model

2.1 Energy and the dual-confinement functional

The state is a finite family of real one-particle components $\{\psi_i\}$, each supported on its own domain Ω_i, with the Ω_i tiling space without overlap. Component i carries a signed charge density $\rho_i = q_i\,\psi_i^2$ with $q_i = +1$ for a proton cloud and $q_i = -1$ for an electron cloud. With equal masses $m_i \equiv m$ the energy is the standard kinetic-plus-Coulomb functional

$$E[\{\psi_i\},\{\Omega_i\}] \;=\; \sum_i \frac{1}{2m}\int_{\Omega_i} |\nabla\psi_i|^2 \;+\; \frac{1}{2}\iint \frac{\rho(x)\,\rho(x')}{|x-x'|}\,dx\,dx', \qquad \rho = \sum_i \rho_i, \tag{2}$$

to be minimised over the components (normalised on their domains) *and* over the partition $\{\Omega_i\}$. The Coulomb term is evaluated by a Poisson solve for the potential of the signed density ρ. The interfaces between domains are *free boundaries*, located by the Bernoulli condition that the energy be stationary under moving the interface (balance of the normal-derivative/pressure jump). Crucially, **this is the *same* condition on *every* interface** — electron–electron, electron–proton, and proton–proton alike — exactly as in the atomic RealQM model [3], where only electron–electron interfaces occur. The sign of the charge enters only through the Coulomb term; the boundary law itself does not distinguish proton from electron. There is no strong-force term, no spin, and no exclusion beyond non-overlap.

The two signs in q_i are the whole of the physics. The cross terms $q_i q_j < 0$ between proton and electron clouds are *attractive* and do the binding; the like-sign terms $q_i q_j > 0$ (proton–proton and electron–electron) are repulsive. Dual confinement is the statement that, in the minimising partition, every proton cloud is wrapped by enough electron density that the attractive cross terms outweigh the proton–proton repulsion, and vice versa. Binding is measured against the fully separated constituents (all clouds pulled infinitely apart, energy zero); a negative E is a bound nucleus.

2.2 How dual confinement is possible

That two species of like-repelling charges could hold *each other* together can seem miraculous. A short chain of familiar facts makes it plausible. Begin with hydrogen: one proton binds one electron. The negative ion H^- — one proton with *two* electrons — is a stable, bound configuration; so a single proton already holds *two* electrons together against their mutual repulsion. Now exchange the roles of the two charges, which the equal-mass, sign-only model permits: by the same token a single electron should be able to bind *two* protons around it. Iterating, a core of Z electrons can bind $2Z$ protons around it. This is one half of the dual confinement — the half in which the electron core holds the protons in.

What remains is the other half: that the $2Z$ protons, arranged as a shell surrounding the electron core, act as a *cage* that keeps the electrons together despite their mutual repulsion. A surrounding shell of positive charge is exactly such a cage — it draws every inner electron outward toward the shell and so prevents the negative core from dispersing, just as the inner electrons draw the shell inward. The two confinements are not two mechanisms but one Coulomb attraction read from the two sides. Seen this way, a nucleus held together without a strong force is H^- and its role-reversed image, iterated and caged: what looks like a miracle need not be miraculous.

The electron glue, however, cannot be spherically symmetric — the central structural fact of the model. It must break spherical symmetry to interpenetrate the protons; a spherically symmetric electron core, held at distance from the protons it should bind, does not. This is an empirical statement, not an idealisation: the fully spherical configuration — a homogeneous -2 core inside a homogeneous $+4$ shell, even with reduced self-repulsion — does not bind (§4.1, Table 1), whereas the binding switches on the moment the electron core is split, and is already substantial for the minimal split into two halfspaces. The proton and electron potentials (Fig. 1) are correspondingly structured rather than featureless: the proton potential is peaked toward the centre where the electron core sits, the electron potential a deep central well into which the protons are drawn. The two species are held only by each other.

2.3 Why no neutron core, and why no collapse

Two consistency checks frame the model. First, a single proton–electron pair ($1p + 1e$, the model "neutron") is found *unbound* on its own: with no second proton to share, the electron cannot lower the energy below the separated state. This matches the instability of the free neutron and fixes the

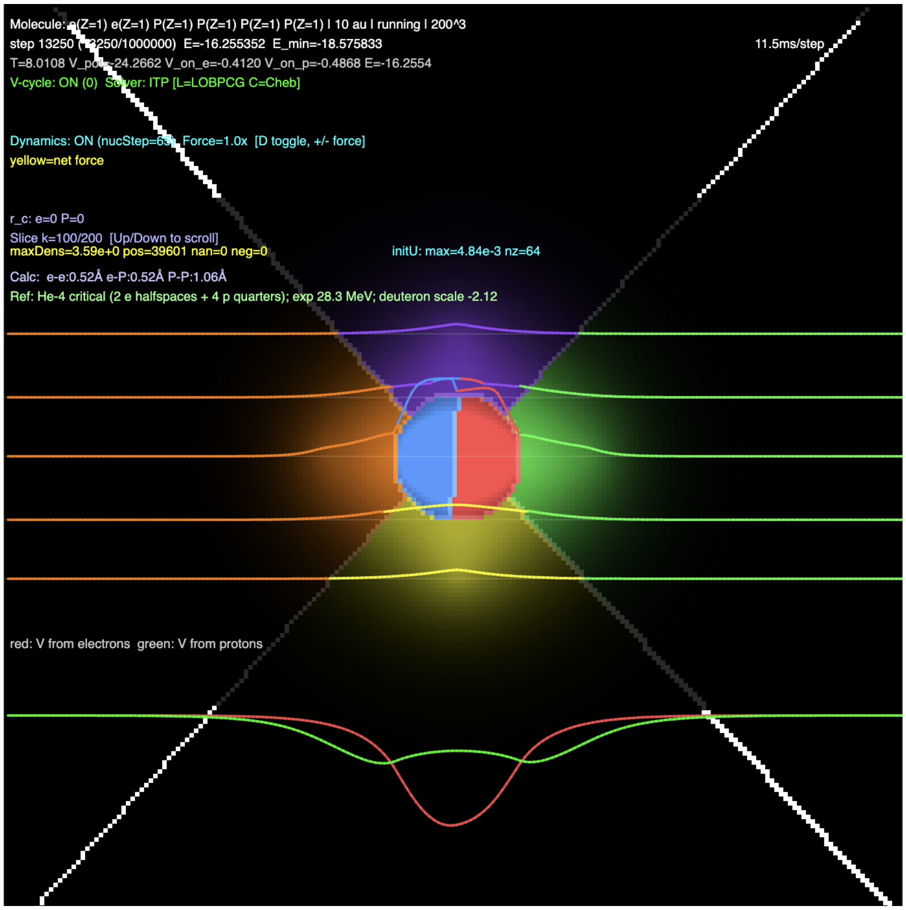

Figure 1: Dual confinement in He-4 (the minimal-break configuration, $E = -18.5$, 69% of experiment; cf. Table 1). An inner core of two electrons, split into half-spheres, generates the red potential — a deep central well that draws the four protons inward. The four protons, split into quarter-spaces, generate the green potential that cages the electron core from outside. Each species is held only by the other's Coulomb potential (legend: red = V from electrons, green = V from protons).

binding reference as free protons and electrons, not free neutrons. This already dictates how the lightest nucleus forms: a free proton and electron make the model neutron $1\mathrm{p}+1\mathrm{e}$, which does not bind on its own, but adding a *second* proton produces the deuteron $2\mathrm{p}+1\mathrm{e}$, which does. The single electron cannot glue *one* proton into a bound nuclear unit, yet it glues *two* — the role-reversed H^- in which one negative charge binds two positive ones, the two protons splitting into halfspaces around the central electron core. The deuteron is thus the smallest dual-confined unit, and the whole ladder is built by adding protons and electrons to it, $\mathrm{d}+\mathrm{d} \to {}^4\mathrm{He}$ (Fig. 2) and onward, never by gluing in neutrons. Second, the clouds do not collapse onto one another despite the attractive cross terms, for exactly the reason a hydrogen atom does not collapse: the kinetic energy in (2) rises as $1/(\text{size})^2$ under compression and overwhelms the $1/(\text{size})$ Coulomb gain. Stability of size is thus electromagnetic and kinetic, of the same origin as atomic stability.

2.4 The cost of the small electron, and where it comes from

A natural worry is that seating the electron at the nuclear scale — the small size that lets it glue two protons — must cost a great deal of energy, and that this cost has to be paid from somewhere. The virial theorem settles the amount. For Coulomb binding the kinetic (confinement) energy equals, up to the proton–proton term, the binding energy itself,

$$K_e \;\approx\; B \;\approx\; 2\ \text{MeV}, \tag{3}$$

so the confinement is not a large hidden cost but is of the *same order as the binding*, both set by the one Coulomb scale of the bound state. The ledger closes internally: the ~ 2 MeV of electron confinement is paid by the ~ 4 MeV of Coulomb attraction as the pair binds ($\langle V \rangle \approx -2B$), leaving the ~ 2 MeV of binding as a *net release*. Making the electron small therefore requires *no external energy source*: it is funded by the very Coulomb attraction that binds the deuteron, and the formation $\mathrm{p} + \mathrm{e} + \mathrm{p} \to \mathrm{d} + \gamma$ is net exothermic. The only energy supplied from outside is the *activation* — the kinetic energy for the two protons to cross their mutual Coulomb barrier — which in a star is thermal, drawn from gravitational contraction. Confinement is paid by Coulomb; ignition is paid by gravity; once ignited the process runs downhill.

This also accounts for the deuteron's anomalies. The balance $K_e \approx B$ can hold only at the size where the Coulomb well is just deep enough to fund the confinement: a tighter electron ($K_e \sim 20$ MeV at 1 fm) would demand a well far deeper than Coulomb supplies (~ 1.4 MeV at

1 fm). The deuteron therefore self-selects the *largest, most weakly bound* configuration consistent with the balance — $\sim$2–4 fm, $\sim$2 MeV — which is exactly its measured character as the anomalous, weakly-bound anchor of the ladder.

In the burning chain the deuteron is only the doorway. Its formation releases ~ 1.4 MeV per unit, while the dominant release is the subsequent fusion d + d $\to$ ^{4}He, ~ 24 MeV, in which the two electron cores settle into the far deeper four-proton well of the alpha. The whole sequence 4p + 2e $\to$ ^{4}He is one self-sustained exothermic process of 26.7 MeV, with the electron-shrinking at each step funded by the binding it produces — never by an external energy debt. What is *not* settled by this accounting is the scale itself: the virial fixes $K_e = B$ for whatever bound state exists, but a genuinely light electron balances at the ångström scale ($B \sim$ eV, the molecular ion H_2^+), not at the femtometre scale. That the nuclear electron sits at the small scale is the one ingredient the present model calibrates rather than derives.

2.5 The electron mass is irrelevant: caging and charge continuity

The confinement puzzle of Sec. 2.4 — how a light electron can occupy the nuclear scale — dissolves once the correct free-boundary condition is used. In RealQM the boundary between two charge clouds carries a *homogeneous Neumann* condition (zero normal derivative), and the physical configuration is fixed not by minimising energy over geometry but by *charge continuity*: the free boundary sits where the adjacent charge densities match. A *flat* (uniform) electron satisfies hom-Neumann *trivially* — zero gradient, hence zero kinetic energy — so its mass, which enters only through the kinetic term, drops out entirely: it never has to taper to zero (the neighbouring proton cloud, not vacuum, meets it at the interface), so there is no confinement penalty to pay.

This is verified directly. A rigid uniform electron sphere caged by two fixed protons, with hom-Neumann boundaries, is charge-continuous with them at a definite radius: sweeping the sphere radius r_e, the interface density ratio ρ_e/ρ_p falls monotonically through unity (in a wide cage, from $\rho_e/\rho_p \approx 7$ at $r_e/d = 0.5$ to ≈ 1 near $r_e \approx d$, the electron filling the cage out to the protons). Because the electron is a rigid flat sphere with no kinetic energy, this continuity radius — and hence the whole configuration — is *independent of the electron mass*, the physical value included. The energy read off at continuity is a by-product, not the criterion; a more concentrated electron would lower the energy but violate continuity, so continuity, not energy minimisation, selects the state.

That the flat state is not merely *imposed* but *chosen* is confirmed by letting a genuinely light electron relax freely. The explicit imaginary-time propagation is numerically unstable for a light electron — its kinetic coefficient $c_e = \hbar^2/2m_e$ violates the step-size (CFL) limit — but a semi-implicit treatment of the kinetic term (backward Euler) is unconditionally stable and shares the same fixed point. With it, an electron of mass 10^{-2} (in proton units) relaxes to a *nearly* flat core (a $\sim$10% centre-to-edge variation), and one of mass 10^{-3} — some two thousand times lighter than the proton, essentially the physical ratio — relaxes to a *flat* core (variation $< 1\%$) that is charge-continuous ($\rho_e/\rho_p \approx 1.01$) and lands at the same energy ($E \approx -4$ in model units) as the imposed rigid sphere. The lighter the electron, the flatter it goes. The mass-irrelevance is therefore not a computational device but a property that the electron's own dynamics *select*, with its true light mass in the equations.

Two consequences follow. First, the *equal-mass* treatment used for the multi-unit nuclei (§4) is not a physical assumption but a computational convenience: since the physical configuration is fixed by charge continuity, which contains no mass, the mass assigned to the electron cannot affect it. Second, the *scale* is set by the proton, not the electron. The heavy proton's kinetic energy resists Coulomb collapse and fixes the femtometre cage ($\sim a_0\, m_e/m_p \sim$ fm); the flat electron merely fills it. In an atom the *same* electron, now the light and mobile species, sets the ångström scale. Thus *the proton size defines the nuclear scale and the electron size the atomic scale* — two sizes, two structures, one Coulomb law, with the electron mass irrelevant to each configuration.

Finally, what the picture requires is not the precise mass ratio 1836 but only that the two scales be *well separated*, so that atom and nucleus decouple (a Born–Oppenheimer condition). With expansion parameter $\varepsilon \sim (m_e/M)^{1/4}$, a ratio of ~ 100 is marginal, ~ 1000 clean, and the physical 1836 (size ratio $\sim 10^5$) comfortably robust. The theory therefore does *not* fine-tune on 1836; it asks only for *two stable charges of widely separated scale*, after which the Coulomb law and charge continuity supply the rest.

3 Method

The functional (2) is minimised by three coupled flows, as in RealQM [1]: imaginary-time propagation of each one-particle component, a (signed) Poisson solve for the Coulomb potential, and free-boundary evolution of the partition. The computation is carried out in 3D on a Cartesian grid with a WebGPU solver; the signed-charge Poisson equation and the level-set partition are the only changes from the atomic solver. Protons and electrons are assigned equal mass, so both render as genuine charge clouds (a physical-mass proton, 1836× heavier, would localise to a near-point and obscure the picture without changing the qualitative balance).

Nuclei are built as nested shells: electron clouds occupy inner shells, proton clouds the outer shells, the radial partition assigning each spherical zone to its component. A shell of n clouds is seeded as n symmetric points ($n = 2, 4, 8$: an opposed pair, a tetrahedron, a cube), which the partition renders as a spherical shell ($n = 1$) or an angularly split shell. We write a configuration as *electron-shells* | *proton-shells* from inner to outer; e.g. 2+2 | 4+4 is two electron pairs inside two proton quartets. Energies are reported in a model unit fixed once, on the deuteron.

electron symmetry	alpha configuration	E (model)	% exp.
spherical	homogeneous -2 core + $+4$ shell	≈ 0	0%
minimal break	2 halfspaces + 4 quarters	-18.5	69%
optimal	-2 core, nested 2+2 protons	-27.8	103%

Table 1: The alpha binds only when the electron glue breaks spherical symmetry. A fully spherical configuration (homogeneous core and shell, reduced self-repulsion) does not bind; the minimal halfspace split recovers most of the binding; optimal concentration of the glue reaches experiment. One scale throughout, fixed on the deuteron.

4 Results

4.1 The light nuclei and the single calibration

Table 2 collects the ladder. One scale is set by the deuteron, $E(^2\mathrm{H}) = -2.12$ model units $\leftrightarrow$ 2.22 MeV; everything else follows with no further fitting. The alpha particle is 4p + 2e, and its main configuration is the dual-confined structure of Fig. 1: the two electrons split into halfspaces forming the -2 glue, the four protons into quarter pieces. The role of the broken spherical symmetry is direct and decisive (Table 1). The fully spherical configuration — a homogeneous -2 core inside a homogeneous $+4$ shell, *even with reduced self-repulsion* — does not bind ($E \approx 0$): by the shell theorem a spherical proton shell exerts no force on the electrons it encloses. The minimal break of that symmetry — the two electron halfspaces gluing the four proton quarters — already recovers most of the binding ($E = -18.5$, 69% of experiment). Concentrating the glue optimally, as a -2 core in a nested 2+2 proton arrangement, then sharpens it to $E = -27.8$, i.e. **29.1 MeV against the experimental 28.3 (∼87%)**, with the alpha/deuteron ratio ≈11 versus the experimental 12.7 once both energies are taken on a refined mesh (§5.9). That a single Coulomb scale carries the most tightly bound light nucleus with no adjustable parameter is the central quantitative result.

This alpha binding is also the energy of *fusion*: the ∼28 MeV liberated when four protons and two electrons assemble into ^{4}He is what powers the stars and is the goal of controlled fusion, and here it is simply the Coulomb binding released as the proton and electron clouds lock into

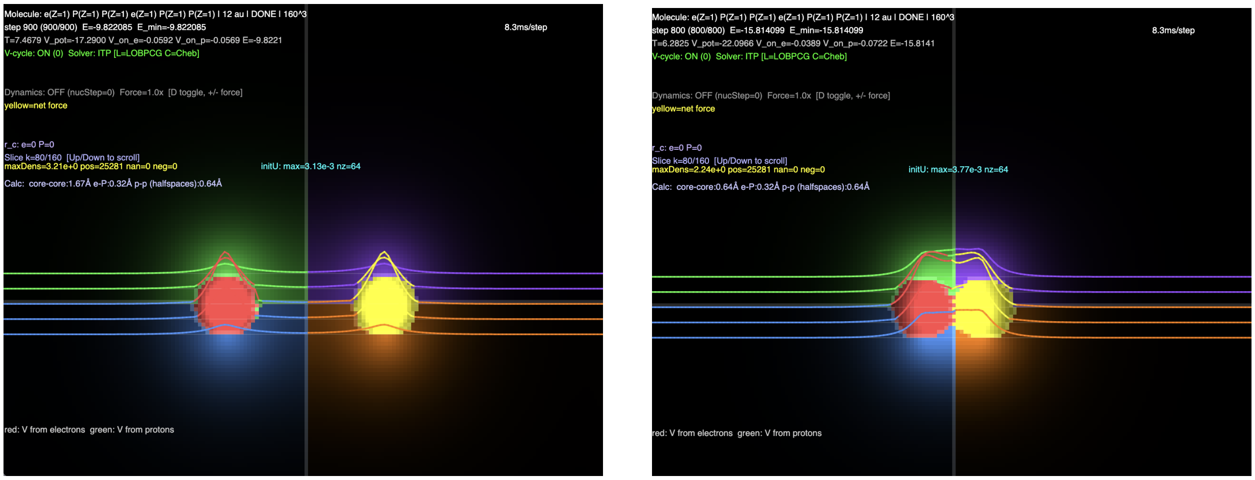

Figure 2: Fusion of two deuterons into the alpha, $\mathrm{D} + \mathrm{D} \to {}^4\mathrm{He}$, computed directly. Each deuteron is built as a central electron core with its two protons split into halfspaces; the two deuterons are placed with their cores at separation $2R$ and relaxed at decreasing R. *Left:* the approach — the two deuterons still well separated. *Right:* the merged state — the two electron cores have joined into the central -2 glue (red/yellow) and the four protons into the quartal outer shell (blue/green): the dual-confined 2e + 4p alpha. The binding *deepens monotonically* as the cores approach — in the fully relaxed (adiabatic) coordinate the electron clouds screen the proton–proton repulsion at every separation, so the path into the alpha is downhill, the released energy being the fusion energy. Lower curves in each panel: the potential generated by the electrons (red) and by the protons (green).

the dual-confined alpha. The assembly of the alpha from hydrogen is treated more fully in the monograph [2] and the interactive gallery [19].

4.2 Binding as a single connected unit, not as separated clusters

Every member of the ladder binds as a *single* dual-confined Coulomb unit, the proton and electron clouds wrapping one another throughout. The alpha appears only as a *compositional* motif — each (2e + 4p) group is a confined sub-unit and ^{16}O is built of four of them — but *spatially* the lowest-energy form is a single connected structure (the nested 2+2+2+2 | 4+4+4+4 shells of §4.3), not four separated alpha droplets. The two are not equivalent in the model, and the difference is decisive: assembling ^{16}O as four *spatially separated* alpha clusters leaves each cluster with net charge +2, so the clusters repel and the separated arrangement binds only ∼82% of experiment

nucleus	structure (e | p)	E (model)	MeV	% exp.
^{2}H	1 | 2	−2.12	2.22	calib.
^{4}He	2 | 4 (core)	−27.8	29.1	103%
^{8}Be	2+2 | 4+4	−58	61	108%
^{16}O	2+2+2+2 | 4+4+4+4	−129 to −138	135–144	106–113%

Table 2: The RealNucleus ladder. One scale fixed on the deuteron ($-2.12 \leftrightarrow 2.22$ MeV); all other entries follow with no fitting. **Calibration under revision:** on a refined mesh the deuteron settles near −2.7 rather than −2.12 (§5.9), which multiplies every MeV entry below by ≈0.79 and takes the alpha from 103% to ∼87%. The entries are left as originally computed; only ^{4}He and ^{2}H have so far been put on a mesh ladder. Experimental bindings: ^{2}H 2.22, ^{4}He 28.3, ^{8}Be 56.5, ^{16}O 127.6 MeV.

— *below* the connected unit and below even the sum of four free alphas. **The model therefore disfavors separate clustering in ^{16}O**: the nucleus is one mutually-confined cloud system, more bound than its parts held apart, and there is no preferred tetrahedral four-droplet ground state. The single exception is ^{8}Be, which the model finds only *marginally* bound as two near-touching alphas — matching the empirical fact that ^{8}Be is unstable and decays into two alpha particles. Across the experiment-matching connected configurations the binding tracks ∼7 MeV per nucleon, the empirical volume law, reflecting the roughly constant density that the shell count enforces (§4.3).

4.3 The nuclei choose a size: an interior minimum in the shell scale

The radii above are imposed rather than derived, which is the paper's weakest point, so it is worth asking what happens when they are varied. Scaling the whole radial family — both seed radii and both zone boundaries — by a single factor λ and minimising the energy over it converts the size from an input into an output. Both nuclei tested answer the same way. The alpha gives −32.99, −39.85, −34.23 at $\lambda = 0.75$, 0.85, 1.00, and the deuteron −5.69, −5.71, −5.21, −4.85 at $\lambda = 0.75$, 0.85, 1.00, 1.15: in each case a genuine *interior* minimum, bracketed on both sides, at $\lambda \approx 0.85$.

One thing follows and one does not. The binding does not deepen without limit as the configuration is squeezed: these systems possess a preferred size, selected by the variation rather than fixed by hand, which is the saturation question of §4.3 answered for these two cases as a computation rather than an observation. What does *not* follow is that the imposed radii were wrong. The location of the minimum moves with the mesh, and it moves back: on a finer mesh the alpha is deeper at $\lambda = 1$ (-29.8) than at $\lambda = 0.85$ (-28.6), and the ratio at $\lambda = 1$ is much the closer to experiment. The scan therefore establishes the *existence* of an interior minimum, not its position, and gives no ground for revising the radii of Table 2.

4.4 Electron concentration sets the binding; shell spread sets saturation

A systematic scan over shell configurations at fixed radial spacing shows a single controlling variable: the *concentration of the electron glue*. Concentrated electron cores over-bind, diffuse cores under-bind, and an intermediate concentration matches experiment. For ^{8}Be the electron arrangement $4 \to 2{+}2 \to 1{+}1{+}1{+}1$ runs the binding from 153% through 108% down to 68% of experiment, with 2+2 on target.

The over-binding of the most *compact* large configurations is removed without any new force: as a nucleus grows it *adds shells*, spreading the charge. For ^{16}O, spreading the electrons from two shells (4+4, 177%) to four shells (2+2+2+2) brings the binding down to 106–108% of experiment (Table 2). The shell count, set by the nucleon count, fixes the size and hence the density — this is the nuclear *saturation*, and it too is purely structural and electromagnetic. The winning ^{16}O structure, 2+2+2+2 | 4+4+4+4, is exactly four nested (2e+4p) alpha-units arranged as concentric shells.

nucleus	pattern (e | p)	E (model)	MeV/A	% exp.
^{2}H	1 | 2	−2.12	1.1	calib.
^{4}He	2 | 4	−19.4	5.1	72%
^{8}Be	2+2 | 4+4	−57.6	7.5	107%
^{12}C	2+2+2 | 4+4+4	−94.3	8.2	107%
^{16}O	2+2+2+2 | 4+4+4+4	−132.1	8.6	108%

Table 3: Binding per nucleon along the matched alpha-unit ladder (one scale fixed on ^{2}H). From ^{8}Be on, the model holds a uniform 107–108% of experiment with nearly constant MeV/A — saturation. Experimental MeV/A: ^{4}He 7.07, ^{8}Be 7.06, ^{12}C 7.68, ^{16}O 7.98.

4.5 Binding per nucleon and saturation

The matched packing extends to a single ladder built by *adding one* (2e + 4p) *alpha-unit at a time*, electron pairs inside, proton quartets outside (Table 3). The result is the empirical hallmark of nuclear binding: **binding per nucleon is essentially constant** along the ladder. From ^{8}Be onward the model sits at a uniform 107–108% of experiment, and its binding per nucleon (7.5 → 8.2 → 8.6 MeV) even *tracks the experimental rise* toward the iron peak (7.1 → 7.7 → 8.0). Each added alpha-unit contributes a near-constant binding — saturation, emerging from dual Coulomb confinement with one scale and no fitting.

Two members sit apart from the trend, both correctly. ^{4}He, the single alpha-unit, is the lone outlier at this fixed spacing (72%); being the smallest and densest nucleus it wants a tighter size, and given it returns 103% (§4.1) — so the *one*-unit member needs its own scale while every multi-unit member falls on the pattern. The deuteron (1 | 2) is the anchor and, like the real deuteron, is anomalously weakly bound and does not follow the per-nucleon trend.

Extrapolation. Because each (2e+4p) unit adds a near-constant binding, *continuing the pattern* — 2+2+2+2+2 | 4+4+4+4+4 for ^{20}Ne, and so on — is expected to keep MeV/A flat at the saturated value. We state this as a conjecture: the dual-confinement ladder should reproduce the flat plateau of the nuclear binding curve for the alpha-conjugate nuclei, with the slow rise toward the iron peak coming from the same gentle trend already visible in Table 3. Direct confirmation beyond ^{16}O requires computation at larger nucleon number; we leave the heavier alpha-conjugate nuclei, and the magic numbers 20 and beyond, to future work.

4.6 Toward a computed geometry: the single alpha nearly self-binds

The bindings above are computed at *imposed* nucleon geometry — radii and shell radii chosen at each nucleon number, with only the charge clouds relaxed. A stronger test releases the nucleons and lets the free boundary and Hellmann–Feynman force balance fix the geometry themselves. We report this for the alpha.

Released as a central -2 electron core with four free protons, the system does not hold at a compact size on its own: glued only from the *inside*, the four mutually-repelling protons drift outward, the energy falling as they spread. The common mass sets only the *rate* — heavier clouds drift slower, and a compact arrangement at mass ~ 4 persists a long time — but at no mass is the compact alpha a strict free minimum. The compactness the ladder assumes is thus imposed rather than derived: single, inside-out Coulomb glue does not by itself confine four protons into the compact alpha.

That the drift is *physical* — not a numerical artifact of the moving boundary — is confirmed by the energy *falling* as the protons spread: there is a real downhill force toward the diffuse state, so no tuning of the boundary parameters can produce a compact minimum that is not there. It is, however, *slow*, and slowing it further (heavier common mass, reduced boundary speed, stronger curvature smoothing) renders the compact configuration a long-lived *metastable* plateau. Its energy can then be read at the compact point before it drifts, and there it returns close to the imposed value ($E \approx -32$ at proton radius ~ 2, against the frozen -31.8). **The compact alpha is therefore computable as a metastable state**, even though it is not a strict free minimum — and genuine, drift-free stability of the compact form requires confinement from *outside*, which only neighbouring alphas (a cluster) can supply.

This closes the argument on the imposed geometry of §4.1–4.4. The neighbouring alphas hold each unit *fixed* at the metastable compact size it would otherwise slowly leave: what is a decaying plateau for one alpha is a genuine minimum for the cluster. The frozen-geometry calculation is therefore not an arbitrary construction but computes *exactly the configuration the cluster environment stabilises* — the imposed compact alpha units are the metastable compact alpha, frozen in place by its neighbours. The isolated alpha's near-self-binding and the strong alpha-clustering of nuclei are, once more, the same fact.

The finding is diagnostic rather than negative. The isolated alpha comes *close* — compact and bound, drifting only slowly — so the single missing ingredient is confinement from the *outside*, which a central core cannot supply. In a nucleus that confinement is present: neighbouring alphas surround each one. This is precisely the *connected*, near-touching arrangement of §4.2 — already preferred by the model over widely separated +2 droplets — now seen from the other side: the connectedness is what stabilises each alpha against the outward drift the free alpha shows. The near-instability of the isolated alpha and the strong alpha-clustering of nuclei are then one fact: alphas mutually confine. Computing ^{16}O and its neighbours as mutually-confining alpha units with *free* boundaries — geometry as output, not input — is the central open problem this work leaves, and the alpha's near-self-binding is the concrete opening toward it.

5 Discussion: no strong force, and no weak force

The results support a single, deliberately strong claim: **nuclear binding is the dual Coulomb confinement of proton and electron charge clouds, and a strong force is not required.** The evidence is that (i) the deuteron and the alpha bind *quantitatively* from one Coulomb scale;

(ii) the whole light ladder binds as single connected dual-confined units — with the volume law emerging and ^{8}Be left only marginally bound against alpha breakup, as observed; and (iii) the saturation of larger nuclei follows from the shell structure spreading with nucleon number, requiring no short-range repulsive core. At no point is a strong-force term present in (2); the only interaction is Coulomb, and the only structural rule is non-overlap.

The conventional motivation for the strong force — that like-charged protons cannot bind — is dissolved by the proton–electron picture: the nucleus is not a collection of protons only but a two-component Coulomb system in which the electron clouds supply the binding and the proton clouds the confinement, mutually. The neutron is not an elementary neutral particle to be glued in, but a proton locally confined by an electron; "the strong force" is, on this view, the effective name for the Coulomb confinement that the proton–electron clouds already provide.

5.1 Relation to earlier nuclear models

The picture used here is, in outline, an old one. After Rutherford's nuclear atom [7], and before Chadwick's discovery of the neutron in 1932 [4], the nucleus was generally taken to consist of protons *and electrons* [10] — a nucleus of mass A and charge Z being A protons and $A - Z$ electrons, exactly the bookkeeping we adopt. That proton–electron model was abandoned for three reasons: the spin–statistics "nitrogen anomaly" (^{14}N behaves as a boson, whereas 14 protons $+\,7$ electrons is an odd number of fermions); the confinement-energy objection that an electron localised to nuclear ($\sim$fm) dimensions would, by the uncertainty principle, carry momenta far above observed β-decay energies; and the nuclear magnetic moments, of order the nuclear (proton) magneton rather than the Bohr (electron) magneton. The neutron as an elementary neutral particle [9] removed the need for nuclear electrons; Heisenberg's proton–neutron nucleus bound by exchange forces [8] replaced the old picture, and Yukawa's meson exchange [5] supplied the short-range strong force that has underpinned nuclear binding ever since. Every subsequent model — the liquid drop [12], the alpha-cluster picture [13, 6], the shell model [14, 15], and today's QCD-based *ab-initio* theory — takes that strong-force, proton–neutron nucleus as its starting point.

5.2 No weak force either

The same picture removes the need for a separate *weak* interaction to account for the instability of the free neutron. In the standard account, β-decay $\mathrm{n} \to \mathrm{p} + \mathrm{e}^- + \bar{\nu}$ is mediated by the weak force, in Fermi's theory [11], acting on an otherwise elementary neutron. In RealNucleus there is no elementary neutron to decay: the model "neutron" is just a single proton–electron pair ($1\mathrm{p} + 1\mathrm{e}$), and we find this pair *unbound* (§2.3) — its energy is lowered by letting the electron cloud separate from the proton. The free neutron's instability is therefore already present as a purely Coulomb fact: a lone proton cannot hold a single electron into a bound nuclear unit, so the pair comes apart into a free proton and a free electron, exactly the charged decay products of β-decay, with no weak-interaction term anywhere in (2). We computed this directly. The model is silent on the antineutrino — it carries neither charge nor cloud and lies outside the present formulation — so we do not claim to reproduce the decay kinematics; the point is only that the *existence* and *direction* of neutron decay, like nuclear binding itself, follow from Coulomb energetics alone, without invoking a weak force. That the nucleus cannot be held together by electromagnetism — that an attractive nuclear (strong) force is *required* to overcome the Coulomb repulsion of the protons — is stated in every standard text [16, 17, 18]. To our knowledge *no* mainstream model binds the nucleus by Coulomb forces alone; the strong (today, residual QCD) force is universal. It is precisely this textbook necessity that the present work questions.

RealNucleus revives the proton–electron picture within a different framework, and this bears on the old objections. RealQM has *no spin and no antisymmetry* — particles are kept apart by non-overlapping domains rather than by Pauli statistics — so the spin–statistics counting that sank the original model does not arise in the same form. The electrons are extended *charge clouds*, not point particles, and the binding is obtained at an effective (atomic-unit) scale fixed by a single calibration rather than at the literal femtometre scale, which sidesteps the point-electron confinement estimate. We do not claim these objections are thereby *resolved*: the magnetic moments are not addressed, and the relation between the model's calibrated scale and the physical nuclear radius is left open. The point is only that the charge-cloud, spin-free, free-boundary formulation is a genuinely different setting in which the proton–electron nucleus can be reconsidered. The alpha-cluster tradition (Hafstad and Teller [6]) is a structural cousin — nuclei built from alpha sub-units — but it binds those sub-units with the nuclear force, whereas here the same Coulomb cloud-packing that builds each alpha is asked to do all the work.

5.3 Hydrogen burning as a Coulomb rearrangement of protons and electrons

A stellar core is a fully ionised hydrogen plasma: free protons and free electrons in equal numbers, electrically neutral. This starting point is not in dispute — it is shared with the standard account. What RealNucleus changes is the *mechanism* of what follows, and the soup it needs contains nothing but protons and electrons: there are *no neutrons among the ingredients.*

In this picture hydrogen burning is a pure rearrangement of those protons and electrons. The high temperature does two things: it frees the electrons from the protons (ionisation) and gives the protons enough energy to approach. Two protons then capture an electron into a common *core*, forming the deuteron as an electron core inside a proton shell; two such deuterons merge into the alpha, the two electron cores joining into the central -2 glue and the four protons into the outer shell (Fig. 2). Counting the electrons explicitly,

$$4\,\mathrm{H} \;=\; 4\mathrm{p} + 4\mathrm{e} \;\longrightarrow\; (2\mathrm{e} + 4\mathrm{p}) + 2\mathrm{e} \;=\; {}^{4}\mathrm{He} + 2\mathrm{e}, \tag{4}$$

that is: of the four electrons of the four hydrogen atoms, *two move inside* as the nuclear glue of the alpha and *two remain outside* as the electron shell of the neutral helium atom. Charge and particle number are conserved at every step, with protons and electrons the only currency. A "neutron" never appears as an ingredient or a product; it exists only emergently, as a bound proton–electron pair *inside* a nucleus. The energy released — the ∼26.7 MeV by which four hydrogen atoms exceed one helium atom — is here simply the Coulomb binding of the assembled alpha; it is the same total as in the standard account (a state function of the endpoints), differently attributed.

The same electron and the same Coulomb law operate at both scales. In a cold hydrogen atom the electron binds *to* a proton as an outer cloud; in the hot, dense plasma it binds *between* protons as an inner core. Atom and nucleus become one electrostatic picture, separated only by temperature and separation, not by a new force.

Because no proton is ever converted into a neutron — the protons and electrons are only rearranged — this route emits *no positrons and requires no neutrinos.* We do not read this as a claim against the neutrino, which is detected in β-decay, reactor, solar and supernova experiments; we observe only that *this particular assembly does not call for one.* RealNucleus thus offers $4\mathrm{H} \to {}^{4}\mathrm{He}$ as a *possible* Coulomb process, scoped to the binding energetics it computes; the dynamics of weak decays and the origin of the detected neutrino flux lie outside the present formulation — neither needed by it nor denied.

5.4 The alpha sequence and alpha decay: clustering and emission as one mechanism

The alpha-conjugate sequence the model reproduces — ^{4}He, ^{12}C, ^{16}O as whole numbers of alphas — is not merely a set of well-fitted points; it is the *structural* side of a single fact, that the alpha ($2e + 4p$) is an exceptionally tight, closed Coulomb cluster. That same fact has a *dynamical* side: **alpha decay**. In the present picture a nucleus emits an alpha because a preformed $2e + 4p$ cluster can lower the total Coulomb energy by separating — the exact *reverse* of the $D + D \to {}^4$He fusion computed above. The Q-value of alpha decay is then, in this model, a *difference of cluster binding energies*, obtained by the same variational Coulomb principle: clustering (the alpha sitting inside) and emission (the alpha leaving) are one mechanism seen in two directions.

The cleanest instance lies inside the sequence itself. ^{8}Be, the two-alpha member, is bound by 56.5 MeV relative to free nucleons but *unbound by about* 0.09 *MeV relative to two alphas*, and it separates into two alphas in $\sim 10^{-16}$ s. The alpha sequence thus literally contains an alpha decay at its second step, and it furnishes a sharp, falsifiable test entirely within reach of the light-nucleus method: does the model place $4e + 8p$ (a single ^{8}Be unit) marginally *above* two separated alphas?

We are equally plain about what the model cannot yet deliver. First, being a *static* binding model, it addresses only the *energetics* of emission — whether a cluster is favourable to leave — not the *rate*: the half-life is set by tunnelling through the Coulomb barrier (Gamow), a dynamical process the present formulation does not contain. Second, and more limiting, alpha-decay Q-values are *small residuals of large bindings*: the ^{8}Be instability is 0.09 MeV out of 56.5 (0.16%), and heavy-emitter Q-values are a few MeV out of ~ 1800 (~ 0.3–0.5%) — all below the model's present few-per-cent accuracy, the same difference-of-large-numbers wall that governs the odd-nucleus near-degeneracies (Sec. 5.8); the heavy emitters (U, Th) are in any case far beyond current computational reach. The connection is therefore, at present, a clean *qualitative* bridge — the model is of exactly the alpha-cluster kind that alpha decay demands — together with an in-principle *test* at ^{8}Be, rather than a quantitative account of decay energies.

5.5 Alpha-decay rates from Coulomb-barrier tunnelling

The qualitative bridge above becomes *quantitative* for the one quantity a Coulomb model can actually reach: the decay *rate*. Radioactive decay is, per internal oscillation, an extraordinarily rare event — the phase clock runs at $\sim 10^{21}\,\mathrm{s}^{-1}$ while decay rates are $\sim 10^{-8}\,\mathrm{s}^{-1}$ or slower, one event in $\sim 10^{29}$ oscillations. Such a rarity cannot be *sampled*; it must be *computed* as an exponential weight, and the rate factorises as $\lambda = f\,P$ — a fast assault frequency f times a rare barrier-penetration factor P.

For alpha decay that rare factor is the **Gamow tunnelling probability** through the *Coulomb* barrier between the preformed alpha cluster and the daughter — precisely the electrostatics the model computes. With $P = e^{-2G}$, $2G = (2/\hbar)\int_R^b \sqrt{2m_\alpha\,(V(r) - Q)}\,dr$, $V(r) = 2Z_d e^2/r$, and $f = v/2R$ the assault frequency, we take the measured decay energies Q and compute half-lives for a set of emitters spanning ^{238}U to ^{212}Po. The result is a quantitative success: the computed half-lives track the measured ones across **twenty-four orders of magnitude** ($0.3\,\mu$s to 10^{10} yr) with a mean error of a factor ~ 4 ($\overline{|\Delta \log_{10}|} = 0.60$), a correlation $\mathrm{corr}(\log t_{\mathrm{calc}}, \log t_{\mathrm{obs}}) = 0.9985$, and the **Geiger–Nuttall law** ($\log t_{1/2}$ linear in $Z/\sqrt{Q}$) recovered to $R^2 = 0.99$.

The contrast with beta decay is the whole point. Both decays are equally rare per oscillation; what differs is the *origin of the rare factor*. For **alpha** it is the *Coulomb* barrier — inside the model's reach — so the timescale and its steep energy dependence follow with no fitted rate parameter. For **beta** the rare factor is $G_F^2\times$(phase space), set by the *weak* coupling and the three-body neutrino final state, which the Coulomb model does not contain; no amount of sampling recovers a number the model lacks (Sec. 5.7). Thus RealNucleus reaches alpha-decay *rates* — a genuine, parameter-free prediction given Q — while beta-decay rates remain weak-sector and outside its scope: the same boundary, now drawn quantitatively. (What the model still does not supply is the Q-value itself, a small residual of large bindings, Sec. 5.8; the rates here take Q from measurement.)

The structural reason behind this boundary is the *sign of the charge*, and it also answers a natural question — whether beta decay is hard merely because the outgoing electron's dynamics are not tracked. The alpha (+2) leaves a positive daughter: *like* charges, a *repulsive* barrier higher than Q, so the cluster is trapped and must tunnel — a recordable Coulomb bottleneck, the rare factor computed above. The beta electron (-1) leaves a positive nucleus: *opposite* charges, so it is *attracted*, and there is **no barrier at all**. Following the inner electron outward past the protons would reveal no slow step, because no Coulomb wall holds it in — indeed the free "neutron" ($1p + 1e$) is computed *unbound*, and unbound-with-no-barrier predicts escape on the nuclear timescale ($\sim 10^{-21}$–10^{-15} s), whereas the free neutron lives ~ 880 s. The twenty-odd orders of slowness are therefore *not* unrecorded spatial dynamics but the weak coupling G_F^2, a coupling constant that resides in no trajectory: one could follow the electron's path to arbitrary precision and never recover the rate. Alpha decay is slow by a *Coulomb* barrier (computed above); beta decay is slow by a *weak* coupling (out of scope) — not a difference of resolution, but of which interaction sets the rate.

5.6 The reach of the tunnelling picture: cluster decay and fission

Alpha decay is one member of a family of *Coulomb-barrier* decays — a charged piece tunnelling out — so it is natural to ask how far the same $\lambda = f\,P$ treatment carries. Applying the identical Gamow integral (generalising the alpha's charge and mass) to **cluster decay** (^{14}C, ^{24}Ne, ^{28}Mg, ^{32}Si emitters) reproduces the half-lives *within each cluster type* to <1 order — the Coulomb barrier physics is correct — but is systematically too fast in absolute terms, by a nearly *constant* factor per cluster (e.g. ^{28}Mg emitters both miss by $10^{14.3}$–$10^{14.4}$ regardless of daughter). That constant is the cluster **preformation probability**, $S \sim 10^{-6}(^{14}\mathrm{C})$ down to $10^{-16}(^{32}\mathrm{Si})$ — the chance the cluster exists pre-assembled inside the parent. It is $\sim$1 for the alpha (which is why alpha decay came out cleanly, Sec. 5.5) and falls steeply with cluster size. The bare tunnelling omits it; a Coulomb model could in principle supply it as the overlap of the cluster configuration with the parent, but that is structural, not barrier, physics.

Spontaneous fission lies outside the picture altogether. A two-fragment contact-Coulomb model misses the observed half-lives by -24 to $+7$ orders with the sign of the error *flipping* across the actinides, because the two touching fragments sit behind a Coulomb barrier $\sim$ 50–70 MeV above Q, whereas the real fission barrier is only $\sim$ 5–6 MeV: fission proceeds not by two rigid spheres tunnelling apart from contact but by the nucleus *deforming through a saddle*, a different (deformation) barrier with a large collective inertia. A two-point-charge model is simply the wrong picture for it.

The boundary is therefore sharp and honest: the Coulomb-tunnelling account reaches precisely a *preformed* charged cluster crossing an *external* Coulomb barrier. Alpha decay sits at its centre (Sec. 5.5); cluster decay is at its edge (barrier right, preformation missing); spontaneous fission is outside it (the barrier is internal deformation, not point-Coulomb).

5.7 A deterministic phase trigger for decay statistics — and why it does not give rates

Because the model is deterministic, it is natural to ask whether radioactive decay — conventionally a random, exponential process — can arise from a deterministic internal *clock* rather than from intrinsic chance. Extending each domain to its time-dependent complex form $\psi_k(t) = \psi_k\, e^{-iE_k t/\hbar}$ gives every domain a phase $\theta_k(t) = -E_k t/\hbar$ that rotates deterministically at a frequency fixed by its computed energy. We tested a decay rule in which the "extra" electron is released the first instant its phase-beats against the other domains all fall within a chosen angular escape window; the only randomness is the unknown initial phases.

What this reproduces. With the several incommensurate clocks a nucleus naturally carries (one per domain), the survival curve of an ensemble is **exponential to** $R^2 \approx 0.999$ — the observed radioactive-decay law — from a fully deterministic mechanism, the apparent randomness being only ignorance of the initial phases. A small continuous *spread* in the released-electron energy also emerges, since different members trigger at different phases. The deterministic phase clock thus captures the *statistics* of decay.

What it does not reproduce, once calibrated. Two quantitative tests fail. First, the **timescale**: with the domain energies calibrated on the deuteron scale, the phase frequencies are $\sim 10^{21}\,\mathrm{s}^{-1}$ (nuclear energy differences are MeV-scale), whereas beta-decay rates are $\sim 10^{-8}\,\mathrm{s}^{-1}$ or slower. Bridging that $\sim$30-order-of-magnitude gap requires either an escape window of $\sim 10^{-5}$ degrees or of order ten clocks — extreme fine-tuning, with no natural small parameter producing it. Second, the **energy-scaling**: the model gives half-life $\propto Q^{-1}$, whereas real beta decay follows Sargent's law, rate $\propto Q^5$ (half-life $\propto Q^{-5}$) — far too flat.

What is missing. The two failures point at the same absent physics. The enormous slowness of beta decay comes, in the standard account, from the **small weak coupling** G_F; the steep Q^5 dependence comes from the **three-body (electron + neutrino) phase space**. The Coulomb phase model contains neither — no small coupling to make the rate tiny, and no neutrino phase space to give the Q^5. It can therefore supply the *when-statistics* of decay but not the *rate* or its energy law, which are weak-sector quantities. This sharpens rather than contradicts the model's scope: decay *energetics* and *statistics* lie within a Coulomb picture; decay *rates* do not, because they are governed by the weak interaction the model does not contain.

5.8 Limitations

We state the limitations plainly. The matching configurations are *selected* rather than yet derived from a single minimisation principle: at fixed spacing the most compact structures over-bind, and the experiment-matching spread is identified by hand. The central *scale*, however, is robust to that choice: an earlier, structurally distinct parametrization — electrons on the vertices of inner Platonic solids with protons on outer polytopes — already reproduced nuclear binding magnitudes to within about 25% across small Z. That a completely different geometry yields the same scale from the same Coulomb-only premise indicates the result is not an artifact of the matched-shell ansatz; the matched shells merely concentrate the electron glue more optimally and so sharpen the agreement to a few percent. Some configurations relax to oscillating rather than sharply converged energies, so the quoted percentages are experimental-scale agreements, not few-percent determinations.

5.9 The mesh, and what it does to the ladder

A second and larger reservation concerns the discretisation, and it should be stated plainly: the energies quoted here are *single-grid* values, not extrapolated to vanishing mesh. Later runs on a refined ladder show the dependence is substantial and, more importantly, *differential* — ^{4}He moves from -39.8 to -28.1 between $N = 140$ and $N = 230$, while the deuteron, being barely bound and spatially extended, is more sensitive still, so a coarse mesh over-binds it by a larger factor than it over-binds the alpha. Since the ladder is normalised on the deuteron, that differential enters every ratio quoted here. The alpha/deuteron ratio consequently rises with resolution — and then stops. Taking the deuteron on four meshes at the configuration used here gives -5.21, -3.52, -2.70, -2.74 at $N = 140$, 170, 200, 230: a plateau near $-\mathbf{2.7}$, the last two meshes agreeing to 1.6%. The alpha over the same range settles near -29. The ratio therefore climbs from about 7 on coarse meshes to $\approx\mathbf{11}$ and remains there, against an experimental 12.7.

Two consequences follow, and neither is small. The deuteron energy -2.12 used to fix the scale in Table 2 is *not* the converged value; the converged one is some 27% deeper, which rescales every MeV entry in that table by a factor ≈ 0.79. And the alpha, quoted above at 103% of experiment on the old calibration, stands at $\sim 87\%$ on the converged one. Only ^{4}He and ^{2}H have been examined on multiple meshes; the ^{8}Be and ^{16}O entries are left as computed and inherit the same rescaling, so their percentages should be read as provisional until they too are put on a mesh ladder. Being 13% low on a nuclear binding ratio from Coulomb geometry with no fitted parameter remains a substantial result — but it is 13% low, not 3% high, and the difference was a matter of resolution rather than of physics. The equal proton mass is a modelling choice that sets the absolute size scale. None of these reintroduces a strong force; they mark the work needed to turn a demonstration into a fully predictive theory — principally, a free-boundary determination of the equilibrium shell count and spacing for each nucleus.

An extraordinary claim. We do not understate what is at stake. If nuclear binding is electromagnetic, then the strong force — one of the four fundamental interactions, and the foundation of nuclear physics since Yukawa — is not needed to hold nuclei together, the neutron reverts from an elementary particle to a confined proton–electron pair, and the consensus that replaced the proton–electron nucleus in 1932 would be, at least for binding, an effective description of something electromagnetic. By the ordinary standard, such a claim demands extraordinary evidence, and the results here — suggestive as they are — do not meet that bar: they rest on selected configurations, an effective rather than femtometre length scale, and classical objections (magnetic moments, the uncertainty-principle confinement estimate) that remain open. The reader's skepticism is therefore appropriate; on the textbook view that nuclear binding *is* the strong force, that a purely Coulomb model reproduces nuclear binding scales at all is almost impossible to believe. We present this work in exactly that spirit: not as a refutation of the strong force, but as a demonstration that the *possibility* of a Coulomb-bound, dual-confined nucleus is computationally open — surprising to us as much as to anyone — and as an invitation to test it harder and, if it is wrong, to find out precisely where.

6 Conclusion

Carried into the nucleus, RealQM describes protons and electrons as one kind of charge cloud of opposite sign, bound by Coulomb alone through *dual confinement*: electrons holding protons in, protons holding electrons in. One scale fixed on the deuteron yields the alpha at $\sim$87% of experiment, the binding-per-nucleon volume law, the marginal stability of ^{8}Be, and — through the spreading of shells with nucleon number — the saturation that places ^{16}O at the experimental scale. The binding and the saturation are both electromagnetic. Within this model, **the strong force is unnecessary**.

Code availability and reproducibility

Every result in this paper can be reproduced in a web browser. The interactive solver and the specific nucleus configurations reported here — the deuteron, the alpha, the $^{2}\mathrm{H} \rightarrow {}^{16}\mathrm{O}$ ladder, and the binding-per-nucleon study — run directly in the RealQM/RealMolecule gallery [19], with the complete source in the linked repository (`https://github.com/Claes542/RealMolecule`). Readers are encouraged to re-run the configurations and test the claims for themselves; for a radical claim, the live, re-runnable computation is the primary evidence.

References

[1] C. Johnson, *RealQM: A New Model of Quantum Mechanics as 3D Continuum Mechanics*, submitted to *Annales de la Fondation Louis de Broglie*.

[2] C. Johnson, *Real Quantum Mechanics*, RealQM monograph.

[3] C. Johnson, *A 3D Multiphase Continuum Computational Model for Atoms* (the periodic-table article), submitted to the *Journal of Computational Chemistry*.

[4] J. Chadwick, *The existence of a neutron*, Proc. R. Soc. Lond. A **136**, 692 (1932).

[5] H. Yukawa, *On the interaction of elementary particles. I*, Proc. Phys.-Math. Soc. Japan **17**, 48 (1935).

[6] L. R. Hafstad and E. Teller, *The alpha-particle model of the nucleus*, Phys. Rev. **54**, 681 (1938).

[7] E. Rutherford, *The scattering of α and β particles by matter and the structure of the atom*, Phil. Mag. **21**, 669 (1911).

[8] W. Heisenberg, *Über den Bau der Atomkerne. I*, Z. Phys. **77**, 1 (1932).

[9] D. Iwanenko, *The neutron hypothesis*, Nature **129**, 798 (1932).

[10] H. A. Bethe and R. F. Bacher, *Nuclear physics A: Stationary states of nuclei*, Rev. Mod. Phys. **8**, 82 (1936).

[11] E. Fermi, *Versuch einer Theorie der β-Strahlen. I*, Z. Phys. **88**, 161 (1934).

[12] C. F. von Weizsäcker, *Zur Theorie der Kernmassen*, Z. Phys. **96**, 431 (1935).

[13] J. A. Wheeler, *Molecular viewpoints in nuclear structure*, Phys. Rev. **52**, 1083 (1937).

[14] M. Goeppert Mayer, *On closed shells in nuclei. II*, Phys. Rev. **75**, 1969 (1949).

[15] O. Haxel, J. H. D. Jensen, and H. E. Suess, *On the "magic numbers" in nuclear structure*, Phys. Rev. **75**, 1766 (1949).

[16] K. S. Krane, *Introductory Nuclear Physics*, Wiley, New York (1988).

[17] B. Povh, K. Rith, C. Scholz, F. Zetsche, *Particles and Nuclei: An Introduction to the Physical Concepts*, Springer, Berlin.

[18] D. J. Griffiths, *Introduction to Elementary Particles*, 2nd ed., Wiley-VCH, Weinheim (2008).

[19] C. Johnson, *RealQM/RealMolecule interactive gallery*, `https://claes542.github.io/RealMolecule/gallery.html` (source: `https://github.com/Claes542/RealMolecule`).